

Module 12 - UI System, States & Responsive UX

Smooth MUI-based product experience

SOURCE OF TRUTH: This resource intentionally uses plain JavaScript. The locked stack is React + JavaScript + Firebase + Material UI (MUI). Do not introduce Next.js, TypeScript, shadcn/ui, Redux, or another backend framework unless the project owner explicitly changes this specification.

Scope

Standardize the interface so every feature feels like one product. Material UI is the primary component system; Tailwind may supplement layout/spacing, but do not create duplicate button/dialog/form primitives without need.

Design direction

- Clean SaaS dashboard aesthetic
- High information density without clutter
- Rounded cards, subtle elevation, strong spacing rhythm
- Clear typography hierarchy
- Consistent primary/secondary/danger actions
- Use MUI theme tokens rather than scattered hard-coded values

Required reusable primitives

AppShell
PageHeader
SectionCard
EmptyState
LoadingState
ErrorState
ConfirmDialog
FormDialog
StatusChip
MoneyAmount
AvatarStack
MemberAvatar
ActivityItem
ResponsiveTable

State matrix

State	Expected UI
Loading	Skeleton or progress; preserve layout
Empty	Illustration/icon + explanation + next action
Submitting	Disable relevant controls + progress
Optimistic success	Immediate visual update; reconcile after server response
Error	Actionable message + retry
Forbidden	Do not show destructive controls; show permission-aware message when relevant
Offline/network	Explain that the change could not be confirmed

Forms

- Inline field validation
- Submit-level validation
- Prevent duplicate submit
- Preserve entered values on recoverable errors
- Use accessible labels and focus handling
- Date/time inputs must respect locale but persist normalized timestamps

Responsive rules

- Desktop-first dashboard is acceptable, but every screen must work at 360px width.
- Tables become cards or horizontally scrollable with visible affordance.
- Sidebar becomes temporary drawer on mobile.
- Dialogs should become full-screen or near-full-screen on mobile when content is complex.

Acceptance criteria

- No broken layout at 360px, 768px, 1280px.
- All buttons have clear labels/icons with tooltips when icon-only.
- Loading and empty states feel intentional, not like missing content.
- Focus state is visible for keyboard navigation.